

Topic 2A: Landscape Systems, Processes & Change

— Glaciated Landscapes

Topic 2 offers a choice: **2A Glaciated Landscapes and Change** or **2B Coastal Landscapes and Change**. You answer questions on one of them in Section B of Paper 1. These notes cover 2A. Study must include landscapes from *inside and outside the UK* — Edexcel expects both a relict British upland and an active glaciated environment abroad.

2A.1 Climate Change and the Glacial Record

Cryosphere: The frozen water component of the Earth system — ice sheets, ice caps, glaciers, ice shelves, sea ice, snow cover and permafrost. It covers roughly 10% of the land surface and holds about 68% of global freshwater. It regulates global systems by controlling planetary albedo, driving thermohaline circulation through cold dense meltwater, storing carbon in permafrost, and supplying meltwater to the rivers of billions of people.

Glaciated landscapes are the product of repeated climatic oscillation, not one ice age. The **Pleistocene** began ~2.58 million years ago and contained more than 50 glacial–interglacial cycles, each overprinting and partly erasing the landforms of the last. That is why the landform record is incomplete: you read the most recent glaciation through the fragments of earlier ones.

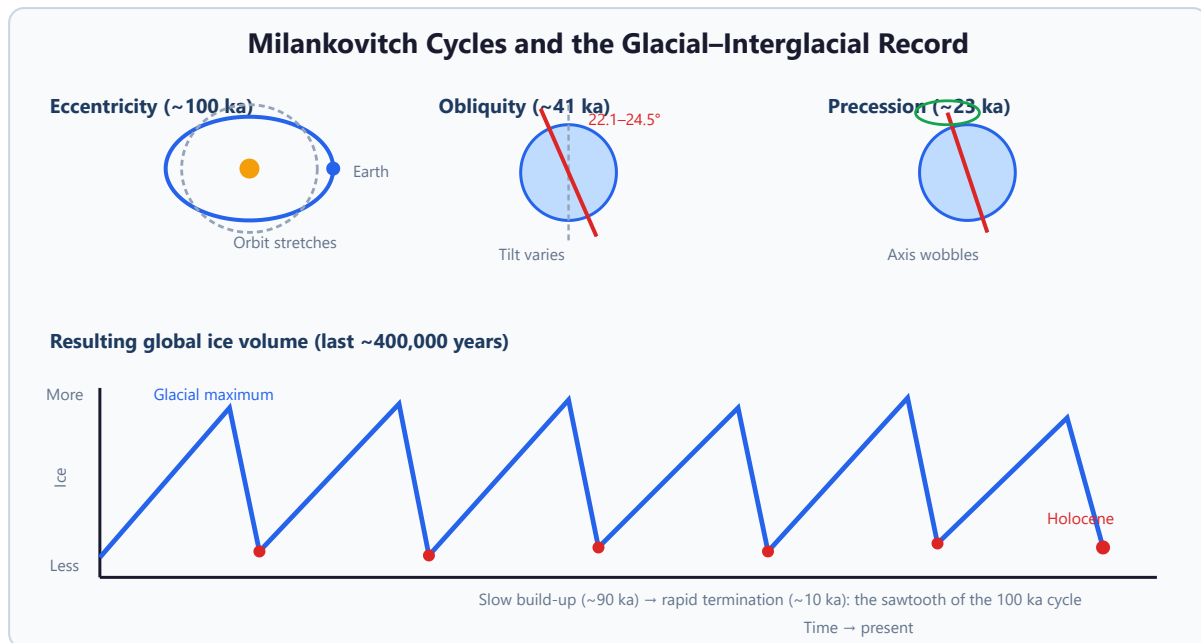
Period	Timing	Characteristics
Pleistocene	2.58 Ma – 11,700 yr BP	Repeated glacial/interglacial cycles; global mean temperature swings of 5–6°C. Ice covered up to ~30% of the land surface at glacial maxima, against ~10% today.
Anglian glaciation	~478,000 – 424,000 yr BP	The most extensive British glaciation; ice reached the outskirts of London and diverted the Thames to its present course.

Devensian glaciation	~115,000 – 11,700 yr BP	The last British glaciation. At the Last Glacial Maximum (~21,000 yr BP) ice reached Norfolk and the Severn estuary; global sea level was 120–130 m lower.
Loch Lomond Stadial	~12,900 – 11,700 yr BP	A sharp cold snap after initial deglaciation; small corrie and valley glaciers re-formed in Scotland, Snowdonia and the Lake District, sharpening many British corries.
Holocene	11,700 yr BP – present	The current interglacial. Warm and comparatively stable, with minor oscillations — the Little Ice Age (c.1300–1850) brought Alpine glaciers to their post-glacial maxima.
Anthropocene	Proposed; not formally ratified	The idea that human activity now dominates Earth surface processes. The International Union of Geological Sciences rejected a formal Anthropocene epoch in March 2024, so it remains informal — but the forcing it describes is real: glacier loss since 2000 cannot be explained by orbital or solar variation.

Milankovitch Cycles

Milutin Milanković showed that the rhythm of glacials and interglacials matches predictable variations in Earth's orbit and axis. These cycles barely change total solar energy received; they change its *distribution* by latitude and season. The critical variable is summer insolation at ~65°N — if northern summers are too cool to melt the previous winter's snow, ice sheets grow.

Cycle	Period	Mechanism
Eccentricity	~100,000 years	The orbit stretches from near-circular to mildly elliptical, altering the perihelion–aphelion difference. Matches the dominant 100 ka rhythm of the late Pleistocene.
Obliquity (axial tilt)	~41,000 years	Tilt varies between 22.1° and 24.5° (currently 23.4°). Greater tilt means stronger seasonality and warmer high-latitude summers, favouring melt.
Precession (wobble)	~19,000–23,000 years	The axis wobbles like a spinning top, changing which hemisphere faces the Sun at perihelion.



Other Causes of Climate Change

- **Variations in solar output:** the Maunder Minimum (c.1645–1715), a prolonged period of very few sunspots, coincides with the coldest phase of the Little Ice Age. Measured solar irradiance has been flat or slightly falling since 1980 while temperatures rise, so solar variation cannot explain recent warming.
- **Atmospheric gas composition:** ice cores from Vostok and EPICA Dome C show CO₂ tracking temperature through the last 800,000 years, oscillating between ~180 ppm in glacials and ~280 ppm in interglacials. It has now passed 420 ppm — outside the entire Pleistocene range.
- **Volcanic eruptions:** large explosive eruptions inject sulphate aerosols into the stratosphere, cooling the surface for 1–3 years. Tambora (1815, VEI 7) produced the "year without a summer" in 1816. Volcanism drives short-term cooling, not glacial cycles.

Exam Tip: Edexcel wants the distinction between a *trigger* and an *amplifier*. Milankovitch cycles are the pacemaker — they set the timing — but orbital forcing alone is far too weak to produce a 5–6°C swing. Ice–albedo feedback and CO₂ released from or absorbed by the oceans amplify the signal. "Milankovitch cycles cause ice ages" is worth far less than explaining the feedback chain.

2A.2 The Distribution of Ice: Past and Present

Ice mass	Scale	Characteristics and examples
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Ice sheet	>50,000 km ²	Continental scale; buries the underlying relief. Only two today: Antarctica (~14 million km ² , ~58 m of sea-level equivalent) and Greenland (~1.7 million km ² , ~7.4 m).
Ice cap	<50,000 km ²	Dome-shaped, submerges upland relief but not a continent. Vatnajökull, Iceland (~7,700 km ²).
Ice field	Variable	Upland ice through which the highest peaks (nunataks) protrude. Southern Patagonian Ice Field.
Valley glacier	Tens of km ²	Confined by valley walls, fed by an ice field or cirque. Mer de Glace; Aletsch (~20 km, largest in the Alps).
Cirque (corrie) glacier	<1—a few km ²	Small ice body in an armchair hollow; often the last ice remaining as a system shrinks.
Ice shelf	Large	Floating extension of an ice sheet that buttresses the grounded ice behind it. Larsen B collapsed in 2002 and the glaciers behind it accelerated up to eightfold.

Polar and Temperate Glaciers

Why thermal regime matters: whether a glacier is frozen to its bed or separated from it by a film of meltwater determines how fast it moves and how much work it does on the landscape. It is the single most useful contrast in the topic.

Feature	Polar (cold-based)	Temperate (warm-based)
Basal temperature	Below pressure melting point; frozen to the bed	At pressure melting point; meltwater present at the base
Location	High latitude — Antarctica, most of Greenland	High altitude, mid-latitude — Alps, Rockies, Southern Alps

Movement	Very slow, often <5 m/yr; almost entirely internal deformation	20–200 m/yr and far more in outlet glaciers; dominated by basal slip
Erosional power	Low — little abrasion or plucking, because the ice cannot slide	High — abrasion, plucking and meltwater evacuation all operate
Meltwater	Almost none; ablation by sublimation and calving	Abundant and seasonal; large fluvioglacial landform assemblage

Most real ice masses are **polythermal** — cold-based at thin margins and beneath the coldest interior, warm-based where the ice is thick enough for pressure and geothermal heat to bring the base to melting point. This explains why erosion is patchy within a single glaciated upland: it is not the presence of ice that erodes, but the presence of *sliding* ice.

Past and Present Distribution

- **Today:** Antarctica and Greenland hold about 99% of the world's land ice. Beyond them, glaciers cover roughly 700,000 km² across more than 200,000 separate ice bodies — Himalaya–Karakoram, Andes, Alps, Rockies, Caucasus. Ice can exist at any latitude given altitude: the Rwenzori glaciers sit almost on the Equator.
- **Evidence for Pleistocene extent:** terminal moraines marking maximum advance (the Cromer Ridge, north Norfolk); erratics traceable to a distant source (Shap granite carried south into Lancashire and the Midlands); drumlin fields; British Geological Survey glacial drift maps; striations recording flow direction; buried tunnel valleys under East Anglia.
- **Relict landscapes:** Snowdonia, the Lake District and the Cairngorms carry corries, arêtes, troughs and ribbon lakes but hold no ice. They are being modified by fluvial, weathering and mass-movement processes — which is why corrie back-walls now sit behind scree.

Case Study: The Lake District — A Relict Glaciated Landscape in the UK

Extensively glaciated during the Devensian and re-glaciated by small corrie glaciers in the Loch Lomond Stadial, the Lake District's radial drainage reflects an ice dome over the central fells from which glaciers flowed outwards down pre-existing river valleys. **Erosional evidence:** Red Tarn beneath Helvellyn and Blea Water beneath High Street are corries with rock lips and tarns; Striding Edge and Swirral Edge are arêtes; Wasdale, Borrowdale and Great Langdale are troughs with truncated spurs and hanging valleys (Sour Milk Gill falls into Easedale). **Ribbon lakes:** Windermere

(England's longest) and Wastwater (79 m, England's deepest) occupy over-deepened trough floors where ice eroded faster across weaker rock or beneath thickened, confluent ice. **Depositional evidence:** recessional moraines across valley floors, drumlins around Kendal, erratics of Borrowdale Volcanic rock well beyond their source. Protected as a National Park (1951) and inscribed as a UNESCO World Heritage Site in 2017 — notably as a *cultural* landscape, recognising that hill farming and the Picturesque movement shaped its present character as much as the ice did.

2A.3 Periglacial Processes and Landscapes

Permafrost: Ground that remains at or below 0°C for at least two consecutive years. It underlies around a quarter of the Northern Hemisphere land surface and is classified by continuity: **continuous** (>90% of ground area, mean annual air temperature below about –6°C; the deepest recorded in northern Siberia exceeds 1,600 m), **discontinuous** (50–90%, broken by unfrozen taliks beneath lakes and rivers) and **sporadic** (<50%, isolated patches at the southern margin). Above it sits the **active layer**, which thaws each summer to 0.3–4 m. Because water cannot drain through the frozen ground beneath, the active layer becomes waterlogged and mobile — the engine of nearly every periglacial process.

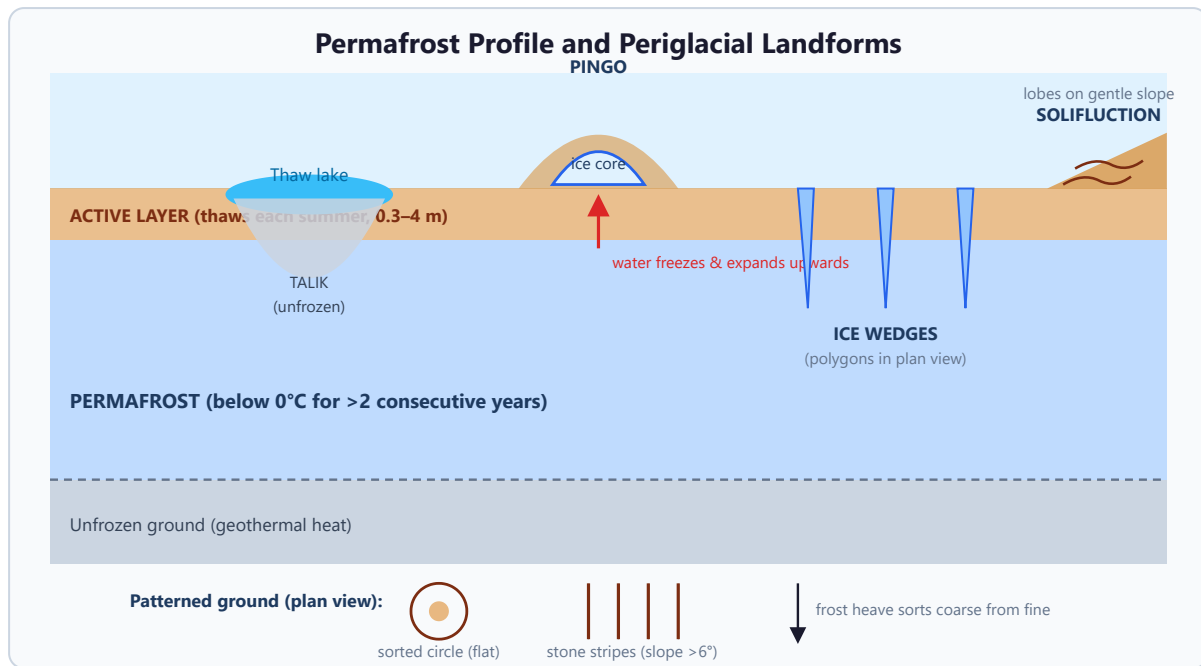
Periglacial Processes

- **Freeze–thaw weathering:** water in joints expands ~9% on freezing and prises rock apart. Needs repeated crossing of 0°C plus moisture. Produces blockfields (felsenmeer) on plateau summits and scree below free faces.
- **Frost heave:** ice lenses in fine sediment push the surface upwards. Stones conduct heat faster than the surrounding soil, so ice forms preferentially beneath them and jacks them to the surface — the mechanism that sorts patterned ground.
- **Nivation:** combined freeze–thaw, chemical weathering and meltwater removal beneath a semi-permanent snow patch, enlarging a hollow into a nivation hollow — the first step towards a corrie.
- **Solifluction:** slow downslope flow of the saturated active layer over impermeable frozen ground, on slopes as gentle as 1–2°. Produces lobes, terraces and unsorted head deposits at slope foot — widespread across southern Britain, which was periglacial rather than glaciated at the Devensian maximum.

- **Meltwater and wind:** the brief summer nival flood carries a disproportionate share of the annual sediment load because the frozen subsurface prevents infiltration. Sparse vegetation and strong katabatic winds allow deflation of silt from outwash plains, deposited downwind as loess.

Periglacial Landforms

Landform	Formation
Ice wedges and polygons	Winter thermal contraction cracks the frozen ground; summer meltwater fills the crack and refreezes, widening the wedge over centuries. In plan the cracks link into polygons 5–30 m across. Fossil ice-wedge casts in southern England prove past permafrost there.
Patterned ground	Frost heave lifts larger stones to the surface, where they roll outwards into sorted circles and polygons on flat ground. On slopes above ~6° gravity drags the pattern downslope into stone stripes.
Pingos	Ice-cored hills 3–70 m high. Open-system (hydraulic) pingos form where groundwater flows under artesian pressure into a talik and freezes. Closed-system (hydrostatic) pingos form where a lake drains or infills, permafrost advances into the saturated sediment beneath, and the trapped water freezes and expands upwards. Collapsed pingos leave circular ramparts — over 100 mapped in the Norfolk Breckland.
Thermokarst	Hummocky terrain of subsidence hollows and thaw lakes formed as ground ice melts. Spreading as Arctic permafrost warms; damages roads, pipelines and buildings and exposes frozen organic carbon to decomposition.
Loess	Wind-blown silt deflated from outwash, deposited in thick, well-sorted, unstratified sheets covering ~10% of the land surface. Highly fertile but extremely erodible once ploughed.



Case Study: The Mackenzie Delta Tundra, Northern Canada

The Mackenzie Delta and Tuktoyaktuk Peninsula hold the world's greatest concentration of pingos — around 1,350 in roughly 16,000 km². Ibyuk Pingo (~49 m high, ~300 m across, about 1,000 years old) is still growing at around 2 cm a year. These are **closed-system pingos**: shallow thaw lakes drained or infilled, permafrost advanced into the saturated sands beneath, and the trapped pore water froze and expanded upwards. The wider landscape shows the full periglacial suite — ice-wedge polygons visible from the air, extensive thermokarst lakes, and solifluction lobes on slopes of only a few degrees. **Change:** permafrost here is warming faster than the global average, and coastal erosion along the Beaufort Sea reaches several metres a year where thawing ice-rich bluffs collapse into the sea; Tuktoyaktuk is relocating buildings away from the shoreline. Ibyuk and neighbouring pingos are protected within the Pingo Canadian Landmark, co-managed with the Inuvialuit under their land claim agreement — protection that treats the landform and the community living with it as one management problem.

2A.4 Mass Balance and Glacier Movement

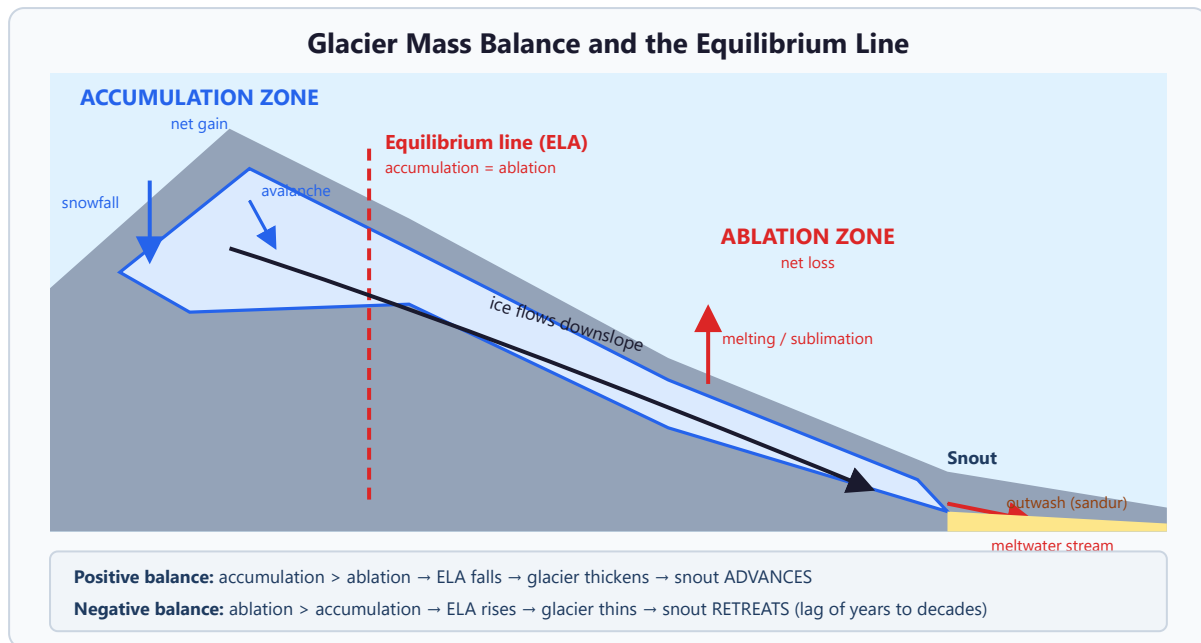
Glacial mass balance (budget): The difference between accumulation (inputs of snow and ice) and ablation (losses) over a year, expressed in metres of water equivalent.

$$\text{Mass balance} = \text{Accumulation} - \text{Ablation}$$

A positive balance means the glacier gains mass and the snout advances; a negative balance means the snout retreats. The snout responds with a **lag** of years to decades, because the signal has to travel the length of the glacier — so a retreating snout tells you about the climate of the recent past, not of this year.

Accumulation (inputs)	Ablation (outputs)
Direct snowfall — the main input in most systems	Melting at the surface, within the ice and at the bed
Avalanches from the valley walls above	Calving of icebergs where the glacier reaches water
Wind deposition of snow redistributed into hollows	Sublimation (ice → vapour), important in cold dry polar air
Rime, hoar frost and refreezing of meltwater	Evaporation from surface meltwater; avalanching off the snout

Equilibrium line altitude (ELA): The altitude at which accumulation exactly balances ablation over a year, separating the upper *accumulation zone* from the lower *ablation zone*. A rising ELA is the clearest single indicator of warming: it shrinks the accumulation area and pushes the glacier into sustained negative balance.



Feedback in the Glacial System

- **Negative feedback:** a thickening glacier flows faster, pushing its snout to lower, warmer altitudes where ablation is higher — removing the extra mass and returning the system towards equilibrium. Heavy debris cover on a snout insulates the ice and slows melting.
- **Positive feedback:** as the **Greenland Ice Sheet** thins, its surface drops into warmer air and melts faster still (the elevation–melt feedback). Meltwater ponds and surface algae lower albedo from ~0.85 for fresh snow to ~0.3–0.4 for bare ice, absorbing more radiation. Surface meltwater drains through moulins to the bed, lubricating basal slip and accelerating outlet glaciers, which then calve more. Greenland has lost mass in most years since the late 1990s, averaging roughly 270 Gt per year over the GRACE satellite era.

How Glaciers Move

Mechanism	Explanation	Where it dominates
Internal deformation (creep)	Ice crystals slip past one another along their basal planes under the weight of overlying ice; the ice behaves as a very viscous fluid. Fastest at depth where pressure is highest.	Cold-based polar glaciers — often their only mechanism
Basal slip	A film of meltwater at the bed reduces friction and lets the whole mass slide over bedrock. Can account for 50–90% of total movement.	Warm-based temperate glaciers

Regelation creep	Increased pressure on the upstream side of a bedrock obstacle lowers the melting point and the ice melts; the meltwater flows round and refreezes in the low-pressure zone downstream, picking up debris. The mechanism behind plucking and roches moutonnées.	Around bed obstacles beneath warm-based ice
Bed deformation	Where a glacier rests on saturated unconsolidated sediment, the sediment itself shears and carries the ice along.	Ice streams and lowland ice sheet margins; thought to build drumlins
Extending and compressing flow	Where the gradient steepens, ice accelerates and stretches, opening transverse crevasses and thinning; where it eases, ice decelerates and thickens, increasing erosion. This alternation is why trough floors are stepped and why rock basins form.	All valley glaciers with irregular long profiles
Surging	Short-lived acceleration to 10–100 times normal velocity, usually from a build-up of subglacial water pressure. Kutiah Glacier in the Karakoram advanced ~12 km in three months in 1953.	A minority of glaciers, but a serious local hazard

Controls on the rate of movement: basal temperature (the biggest single control — warm-based ice can slide, cold-based ice cannot); gradient; ice thickness and mass, which raise basal pressure and lower the pressure melting point; lithology and bed roughness, since permeable or fractured bedrock drains subglacial water and reduces sliding; mass balance, since a strongly positive balance drives faster transfer of ice downglacier; and season — many Alpine glaciers move 20–30% faster in summer when surface meltwater reaches the bed.

Exam Tip: Mass balance questions usually come with data — a table of winter accumulation and summer ablation, or a cumulative balance graph. Calculate the net balance and describe the trend before you explain it. And remember the lag: if asked why a glacier is still retreating after a year of heavy snowfall, the answer is that the snout responds to the cumulative balance of previous decades.

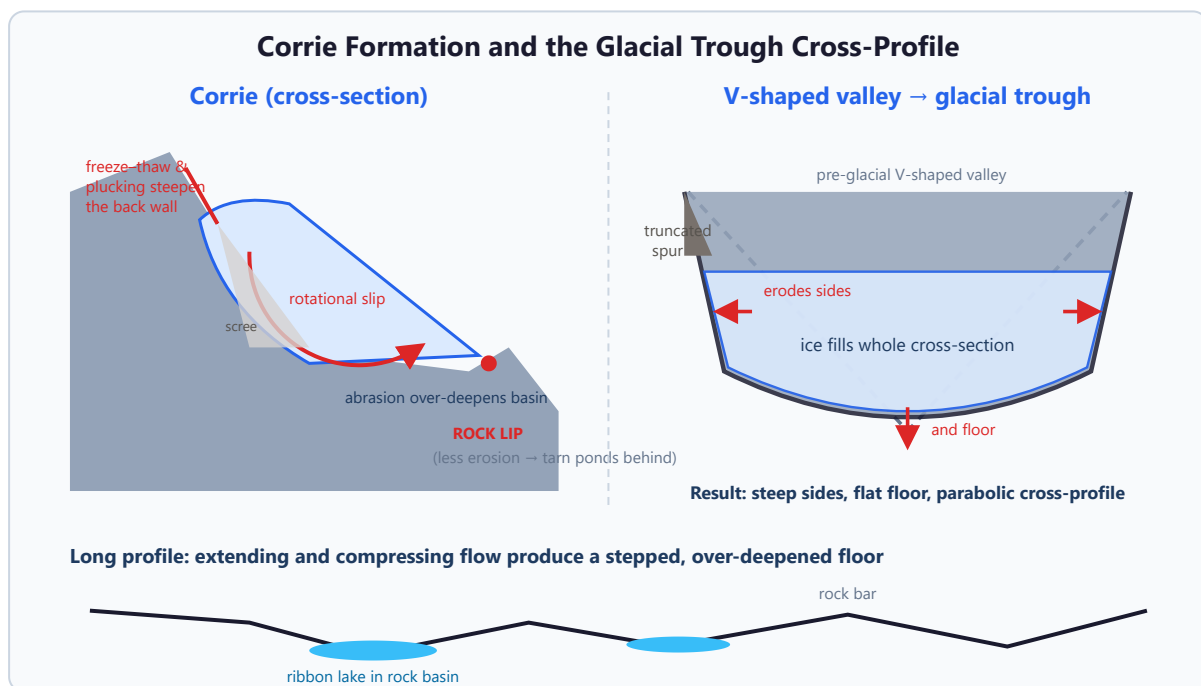
2A.5 Glacial Erosion: Processes and Landforms

- **Abrasion:** debris embedded in the ice is dragged across bedrock, scouring it. Produces striations (recording flow direction), grooves and a fine rock flour that gives meltwater streams their milky turquoise colour. Rate depends on debris concentration and hardness, ice velocity and basal water pressure.
- **Plucking (quarrying):** meltwater refreezes onto jointed or pre-weathered bedrock and pulls blocks away as the ice moves on. Most effective on the lee side of obstacles where regelation refreezing occurs. Produces jagged, stepped surfaces.
- **Crushing and basal melting:** the weight of ice fractures weaker bedrock, preparing it for removal; pressure melting lubricates sliding and chemically weathers carbonate rocks.
- **Subaerial processes above the ice:** freeze–thaw on exposed valley walls supplies angular debris onto the ice surface, and rockfall keeps corrie back walls steep. Not glacial processes — but glaciated landscapes cannot be explained without them.

Entrainment and transport: Debris travels **supraglacially** (on the surface, from rockfall and avalanche), **englacially** (within the ice, after burial by snow or falling into crevasses) and **subglacially** (at the base, from plucking and abrasion). Because ice has effectively unlimited competence, glaciers transport boulders of any size — which is why erratics are diagnostic of ice rather than water.

Landform	Formation	UK example
Corrie (cirque, cwm)	Snow collects in a north- or north-east-facing hollow (shaded and leeward of the westerlies, so snow survives the summer). Nivation enlarges it; ice forms and rotational slip abrades the floor into an over-deepened basin while plucking and freeze–thaw steepen the back wall. Reduced pressure at the downstream edge leaves an unexcavated rock lip , which impounds a tarn.	Red Tarn, Helvellyn; Cwm Idwal
Arête	Knife-edge ridge where two corries erode back-to-back and their back walls intersect.	Striding Edge; Crib Goch
Pyramidal peak	Three or more corries cut back into the same summit, leaving a steep-sided pyramid.	Yr Wyddfa (Snowdon); the Matterhorn globally

Glacial trough	A valley glacier fills a pre-existing V-shaped river valley and erodes floor <i>and</i> sides, leaving a steep-sided, flat-floored trough with a parabolic cross-profile and straightened long profile.	Great Langdale; Nant Ffrancon
Truncated spurs and hanging valleys	Interlocking spurs are cut off square by ice too rigid to wind around them. Tributary glaciers erode less deeply than the trunk glacier, so their floors are left high above it and their streams enter by waterfalls.	Wasdale; Sour Milk Gill into Easedale
Ribbon lake	Long narrow lake in an over-deepened trough section — where rock was weaker, where tributary glaciers joined and thickened the ice, or where compressing flow concentrated erosion. Dammed downstream by a rock bar or moraine.	Windermere; Wastwater; Llyn Ogwen
Fjord	A trough over-deepened below sea level and drowned by post-glacial sea-level rise, often shallowest at the mouth where a rock threshold marks the limit of effective erosion.	Loch Etive; Sognefjord (1,308 m deep)



Landforms of Ice Sheet Scouring

- **Roche moutonnée:** an asymmetric bedrock knoll — smooth and striated on the up-glacier (stoss) side by abrasion, jagged and stepped on the lee side where regelation meltwater refroze and plucked

blocks away. An excellent flow-direction indicator.

- **Knock and lochan:** irregular ice-scoured lowland of rock knolls and water-filled hollows, where an ice sheet exploited a dense joint network. The Assynt coast of north-west Scotland, on Lewisian gneiss, is the classic British example.
- **Crag and tail:** a resistant obstacle (the crag) protects a tapering ridge of softer rock and till on its lee side (the tail). Edinburgh Castle Rock is the crag; the Royal Mile runs down its tail.
- **Differential geology is the control.** Resistant, well-jointed rock is plucked into knolls; weak rock is scoured into basins. The same ice sheet crossing different geology produces completely different landscapes — which is exactly the landscape-systems point Edexcel wants.

2A.6 Glacial and Fluvioglacial Deposition

Till (boulder clay): Material deposited directly by ice. It is *unsorted* (all grain sizes mixed), *unstratified* (no layering) and *angular to sub-angular*, because the clasts have not been rounded by water. Clast long axes tend to align with ice flow — the basis of **till fabric analysis** using rose diagrams.

Lodgement till is plastered onto the bed beneath moving ice (dense, strong fabric); **ablation till** melts out from stagnant ice (looser, weak fabric).

Landform	Formation
Terminal and recessional moraines	A ridge of till marks the furthest advance, built where ablation balanced forward motion long enough to dump debris; further ridges behind it mark still-stands during retreat, and their spacing records the rate of deglaciation. The Cromer Ridge in north Norfolk marks a major ice limit.
Lateral and medial moraines	Lateral moraines are ridges of angular frost-shattered debris along the glacier margins, supplied by rockfall from the valley walls and left perched on the valley sides. Where two glaciers merge, adjacent laterals combine into a medial moraine running down the centre of the ice.
Push moraine	Previously deposited material bulldozed into a ridge by a re-advancing snout; clasts are often tilted up-glacier, which is diagnostic evidence of re-advance.

Drumlin	Streamlined hill of till, typically 5–50 m high and 100–1,000 m long, with a blunt stoss end facing up-glacier and a tapering lee end. Formed beneath actively moving ice, probably by deformation of saturated subglacial sediment. Occurs in "basket of eggs" swarms — the Ribble and Eden valleys, and around Kendal. The elongation ratio (length ÷ width) indicates ice velocity.
Erratic	A rock carried far from its outcrop and identifiable by lithology. Norber erratics (Silurian gritstone on Carboniferous limestone pedestals near Austwick) and Shap granite boulders carried east and south record flow direction and distance.
Till plain	Extensive, gently undulating lowland sheets of till. Much of East Anglia, Cheshire and the Vale of York — agriculturally productive, and the reason the British lowlands are so flat over such varied solid geology.

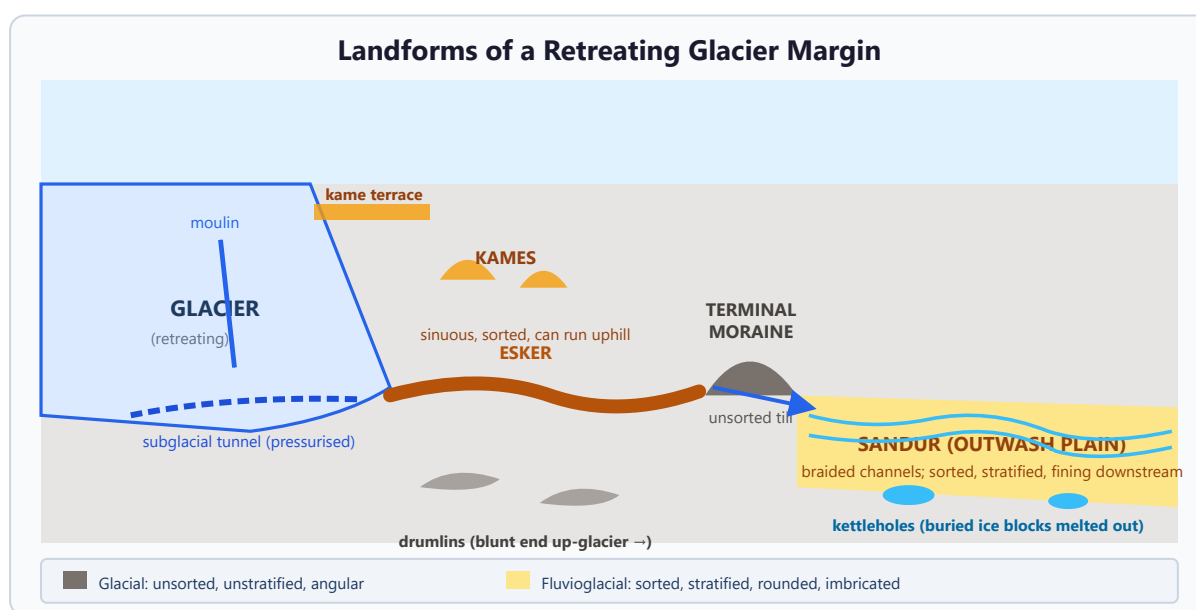
Meltwater and Fluvioglacial Landforms

Meltwater moves **supraglacially** (surface channels feeding moulins), **englacially** (conduits within the ice) and **subglacially** (pressurised tunnels at the bed, which can flow uphill). Discharge is strongly seasonal and often diurnal. Because meltwater's competence and capacity vary with discharge, its deposits differ fundamentally from till.

Property	Glacial (till)	Fluvioglacial (outwash)
Sorting	Unsorted — clay to boulders mixed	Sorted — coarse near the ice, fining with distance
Stratification	Unstratified	Stratified, often cross-bedded, with graded beds
Shape	Angular to sub-angular	Sub-rounded to rounded by attrition in water
Imbrication	None; fabric aligned with ice flow	Clasts stacked overlapping and dipping upstream

Fluvioglacial landform	Formation
Esker	A long sinuous ridge of sorted, stratified sand and gravel deposited in a subglacial or englacial meltwater tunnel, left standing proud when the ice melts. Can run uphill because the water was under pressure.

Kame and kame terrace	Irregular mounds of sorted sand and gravel deposited in supraglacial hollows, crevasses or against the ice margin, let down onto the ground as the ice melts. Kame terraces form along valley sides from streams between the ice and the wall — distinguishable from lateral moraines because they are sorted and stratified.
Sandur (outwash plain)	A broad plain of braided meltwater channels beyond the snout, depositing sorted sand and gravel that fines with distance. Skeiðarársandur in Iceland (~1,300 km ²) is the largest active sandur in the world.
Kettlehole	A depression left where a block of ice buried in outwash later melted and the overlying sediment collapsed; often water-filled.
Proglacial lake and meltwater channel	Lakes impounded between the ice margin and higher ground (Lake Pickering in the Vale of York). When they overtop a col they drain catastrophically, cutting a steep-sided gorge — Newtondale in the North York Moors, far too large for the misfit stream in it today.
Jökulhlaup	A glacial outburst flood, typically from a subglacial lake heated by volcanic activity or a failing moraine dam. The 1996 Grímsvötn jökulhlaup in Iceland peaked at around 45,000 m ³ /s, destroyed sections of the Ring Road and rebuilt the sandur surface in days.



Reconstructing former ice extent. Because each landform records a different aspect of the ice that made it, an assemblage lets geographers reconstruct where ice reached and which way it flowed. Terminal

moraines fix the maximum limit; erratics give provenance, so matching a boulder to its outcrop gives a flow vector of known length (Ailsa Craig microgranite reaches Wales, Ireland and the Isle of Man); drumlin orientation maps flow across whole lowlands; crag and tail, roches moutonnées and striations give local direction; and till fabric analysis distinguishes lodgement till (strong preferred orientation) from ablation till (weak or random).

Exam Tip: The most reliable mark-earner here is the glacial/fluvioglacial distinction. Given a photograph or section diagram of sediment, work the checklist: sorted? stratified? angular or rounded? imbricated? Then name the process and the landform. Examiners repeatedly report that candidates describe the sediment accurately but never say what it *means*.

2A.7 Using and Managing Glaciated Landscapes

The Value of Glaciated and Periglacial Landscapes

- **Environmental and scientific:** ice cores from Antarctica and Greenland provide the 800,000-year record of atmospheric composition on which climate science rests. Glaciated uplands are water towers — the Hindu Kush Himalaya feeds 12 major river systems supporting around 2 billion people, 240 million in the mountains and 1.65 billion downstream (ICIMOD HI-WISE, 2023).
- **Cultural and spiritual:** wilderness value and the Romantic tradition (Wordsworth in the Lake District) created the idea of landscape worth protecting. Many peaks are sacred, and Arctic peoples including the Inuvialuit and the Sámi depend on frozen ground and sea ice for travel and herding.
- **Biodiversity:** tundra supports low-diversity, high-endemism communities; Arctic-alpine plants such as purple saxifrage survive on British corrie walls only as relicts.
- **Economic:** tourism (Eryri receives around 4 million visitors a year, with roughly 600,000 reaching the Snowdon summit), hydroelectric power (Norway generates around 90% of its electricity from hydro, much of it glacier-fed), mining, forestry and hill farming.
- **Role in natural systems:** glaciers regulate river regimes by releasing meltwater in the dry season, and permafrost stores an estimated 1,400–1,600 Gt of organic carbon — roughly twice the amount now in the atmosphere — whose release would amplify warming.

Threats

Threat	Mechanism and impact
Avalanches	Triggered by heavy snowfall, rapid warming, wind slab or human disturbance on slopes of roughly 30–45°. Deforestation of protective forest and off-piste skiing raise risk. The 1999 Galtür avalanche in Austria killed 31 people in a village previously judged safe.
Glacial outburst floods (GLOFs)	Moraine-dammed proglacial lakes grow as glaciers retreat, then fail catastrophically. Risk is rising across the Himalaya and Andes as lake numbers and volumes increase.
Slope instability	Debuttressing as glaciers thin removes support from valley walls, and permafrost thaw in rock faces reduces cohesion. The 2017 Piz Cengalo rock avalanche in Switzerland involved ~3 million m ³ and killed eight hikers.
Tourism and leisure	Footpath erosion, trampling of Arctic-alpine vegetation that may take decades to recover, congestion and second-home inflation of house prices. Ski infrastructure requires forest clearance, ground levelling and artificial snow, which consumes water and energy.
Reservoirs, HEP and urbanisation	Reservoirs drown valley-floor habitat and settlement, trap sediment and alter downstream flow regimes. Valley floors are the only flat land, so settlement and transport concentrate exactly where debris flows, avalanches and floods converge; road-building destabilises slopes.
Climate change	The overarching threat — it amplifies every other one on this list.

Case Study: The Chamonix Valley and Mont Blanc Massif, France — An Active Alpine Valley Under Pressure

Chamonix sits beneath Mont Blanc (4,806 m) in a glacial trough, with a permanent population of around 9,000 and roughly 5 million visitors a year. **Climate change:** the Mer de Glace, France's largest glacier, has retreated more than 2 km since the mid-19th century and thinned by roughly 100–170 m at Montenvers since 1990. The staircase from the Montenvers railway down to the ice cave has been extended repeatedly and now runs to around 580 steps, with dated markers showing where the ice surface stood in 1990, 2000 and 2010 — an unusually legible public record of mass balance. **Hazard:** debuttressing and permafrost thaw have increased rockfall on the Aiguille du Midi and the Goûter route; the Tacconnaz glacier threatens the valley floor with ice avalanches and is protected by one of the largest avalanche-defence structures in Europe; the Tête Rousse glacier holds

a subglacial water pocket that burst in 1892, killing around 175 people at Saint-Gervais, and is now monitored and periodically pumped. **Pressure:** the Mont Blanc road tunnel concentrates freight pollution in a steep-sided trough with poor ventilation, and the Aiguille du Midi cable car delivers very large numbers to a high-altitude site. **Management:** the Espace Mont-Blanc agreement coordinates planning between France, Italy and Switzerland; the summit route is now subject to permit limits after overcrowding incidents; valley traffic restrictions have been introduced. Chamonix shows that in an active glaciated landscape, hazard management, conservation and the tourist economy are one problem, not three.

Case Study: The Himalaya — Climate Change, Mass Balance and the GLOF Hazard

The change: Hindu Kush Himalayan glaciers lost mass around 65% faster in 2010–2019 (0.28 m water equivalent per year) than in 2000–2009 (0.17 m/yr), and ICIMOD's 2023 HI-WISE assessment projects that on the current emissions path they could lose up to 80% of their volume by 2100. Globally, the GlaMBIE assessment published in *Nature* in February 2025 found glaciers lost 273 ± 16 Gt per year between 2000 and 2023 — a 36% increase from the first half of that period to the second — contributing 18 mm to sea-level rise and making glaciers the second-largest contributor after ocean thermal expansion.

The consequence: as glaciers retreat they leave moraine-dammed proglacial lakes. On 4 October 2023 roughly 14.7 million m³ of frozen lateral moraine collapsed into **South Lhonak Lake** in Sikkim, generating an impact wave around 20 m high that breached the moraine and released some 50 million m³ of water into the Teesta valley. The flood destroyed the 1,200 MW Teesta III hydroelectric dam at Chungthang and 13 bridges, affected over 60,000 people across four districts, and killed at least 24 people with more than 70 missing. Analysis published in *Science* attributed it to a multi-hazard cascade combining glacier retreat, permafrost degradation and moraine instability.

Why it matters: this is the specification's "disruption of the hydrological cycle" made concrete — changed meltwater volumes, altered river discharge, enormous sediment yield and degraded water quality — with consequences amplified by human decisions. South Lhonak had been flagged as high-risk and partially drained by siphons before 2023; the mitigation was not sufficient for the magnitude of the failure.

Managing the Threats: A Spectrum of Approaches

Approach	Characteristics	Example
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Total protection	Most human activity excluded; access restricted. Effective for conservation, but excludes local economic use and can be resented.	Antarctic Specially Protected Areas under the 1991 Madrid Protocol
Conservation with managed access	Protection of landform and habitat combined with controlled visitor use — zoning, permits, path hardening, seasonal closures.	Yosemite National Park; Cwm Idwal NNR
Sustainable exploitation	Economic activity permitted within limits designed to keep natural systems functioning; often community-led.	Reduced-stocking hill farming in Eryri; trekking permits in Sagarmatha National Park
Multiple economic use	Landscape managed principally for output — HEP, mining, mass tourism — with mitigation of the worst impacts.	Norwegian glacier-fed hydro schemes; large Alpine ski areas
Exploitation with minimal restraint	Short-term extraction with limited environmental control.	Historic Svalbard coal mining; unregulated road construction in parts of the Himalaya

Players and legislative frameworks. Globally, UNESCO World Heritage designation (the Lake District, 2017; Jungfrau–Aletsch, 2001) and the Antarctic Treaty System's 1991 Environmental Protocol, which bans mineral resource activity other than research. Nationally, national parks (the US National Park Service from 1916; UK National Parks from the 1949 Act), SSSIs and planning controls. Regionally and locally, park authorities managing footpaths and planning policy, municipal avalanche defence and land-use zoning. NGOs such as the National Trust, a major Lake District landowner, lobby, monitor and own land outright. And local communities and indigenous peoples — hill farmers, guides, herders — depend on continued access, which co-management agreements such as the Inuvialuit Final Agreement recognise formally.

Case Study: Yosemite Valley, California — Managing a Relict Glaciated Landscape

Yosemite Valley is a 1,000 m deep glacial trough cut into granite by successive Pleistocene glaciers, with Half Dome and El Capitan as its rock walls and Yosemite Falls (739 m) dropping from a hanging valley. It became a protected reserve in 1864 under the Yosemite Grant — the first time a national government set land aside expressly for preservation and public use — and a national park in 1890.

The pressure: around 4 million visitors a year, historically concentrated in the Valley, which is only about 1% of the park's area. The result was congestion, air quality problems, meadow trampling,

riverbank erosion along the Merced and habituated black bears raiding vehicles. **The response:** the Merced Wild and Scenic River Plan (2014) set user-capacity limits and removed or relocated facilities from the river corridor, restoring around 190 acres of meadow and riparian habitat; free shuttle buses, cycle paths and day-use reservation systems manage arrivals; mandatory bear-proof food storage cut bear incidents sharply; boardwalks keep visitors off fragile meadow. **The tension:** the enabling legislation requires the Park Service both to conserve the scenery and to provide for its enjoyment — two goals that pull against each other — and reservation systems have been criticised as excluding lower-income and spontaneous visitors. Yosemite shows that managing a relict glaciated landscape is mostly about managing *people*: the landforms are not going anywhere on a human timescale, but the ecosystems and the visitor experience are fragile.

Exam Tip: Edexcel's 20-mark questions here almost always turn on *players* and *attitudes*. Structure an evaluation around who wants what: conservationists want preservation; local authorities want employment and tax revenue; residents want affordable housing and reliable access; tourism operators want capacity; farming and indigenous communities want continued use rights; national governments want energy security. A "spectrum of approaches" answer that never names a player is a Level 2 answer at best.

Exam Tip: Think in SYSTEMS, exactly as you would for the coastal option, and always link process to form to time. A glacier is an open system with inputs, outputs, stores and transfers, and most high-tariff questions ask you to trace a change through it: warming raises the ELA → negative mass balance → thinning → debutching and proglacial lake growth → slope failure and GLOF risk → altered river regime and sediment yield downstream. Then place the landform in time — a British corrie was cut by Devensian ice, re-occupied in the Loch Lomond Stadial, and has been modified since by freeze–thaw and scree. Relict landscapes are several process regimes stacked on one another, and saying so separates a top-band answer from a competent one.